

# Umar Turdumambetov

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GitHub | Portfolio

## Professional Summary

Machine Learning and Backend Engineer building production-grade systems for real-time data processing, probabilistic modeling, and decision pipelines. Experienced deploying PyTorch models, designing FastAPI microservices, and optimizing low-latency pipelines for scalable inference. Focused on high-impact systems that convert raw data into reliable, measurable outcomes.

## Technical Skills

**Languages:** Python, C++, Java, JavaScript/TypeScript, SQL (PostgreSQL, MySQL), Bash/Shell, HTML/CSS

**ML/AI Frameworks:** PyTorch, TensorFlow, YOLOv8, LangChain, Scikit-learn, Pandas, NumPy, XGBoost

**Backend & Data Engineering:** FastAPI, Flask, Node.js, Express, GraphQL, Prisma, Redis, Kafka, Airflow, Snowflake

**Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, Kubernetes, GitHub Actions, CI/CD, Linux, REST APIs

**Frontend:** React, Next.js, Redux, TailwindCSS, Vite, Figma

**Tools:** Git/GitHub, Jira, Postman, Vitest, Cypress, Agile/Scrum

## Projects

### Churn Predictor — Production MLOps System | Live on Render

- Built end-to-end ML system on 7,043 customers including data validation, feature engineering, and model deployment, reducing misclassification by 40%.
- Designed FastAPI inference service with CI/CD, Docker, and MLflow tracking, enabling automated and reproducible deployments.
- Implemented monitoring and logging pipelines to track model drift and performance, improving reliability in production.

### PolyPredictor-Kit — 1st Place (QuackHacks 2025)

- Developed real-time prediction pipeline generating sub-200ms probabilistic signals across dynamic queries using async FastAPI and parallel ingestion.
- Processed and structured 3,000+ community data points into Snowflake datasets, improving prediction accuracy by 30%.
- Optimized pipeline throughput and latency for high-frequency inference across multiple active markets.

### PromptLock — LLM Context Compression

- Engineered semantic compression system enforcing strict token budgets (2K–16K) with adaptive prioritization for efficient inference.
- Reduced hallucinations by 60%+ and inference cost by up to 70% across production LLM workloads.
- Improved response reliability by minimizing truncation failures and optimizing context utilization.

### AI Visual Novel Creator — 1st Place (CodeDay 2025)

- Built full-stack AI pipeline integrating LLM narrative generation and image synthesis across 20+ branching story states.
- Reduced generation time by 55% and eliminated 90% of rendering artifacts via caching and GPU optimization.
- Designed scalable architecture supporting dynamic story routing and real-time content updates.

## Experience

### AlgoVerse

Sep. 2025 – Present

AI Researcher (RL & Post-Training)

Remote

- Improved model stability and performance by 15–25% through reinforcement learning and post-training optimization techniques.
- Developed evaluation frameworks and benchmarking pipelines to measure generalization and consistency across tasks.
- Reduced output variance by 20% through systematic fine-tuning and experimental iteration.

### Canva

Nov. 2025 – Dec. 2025

AI Design & Data Storytelling Extern

Remote

- Built modular AI content pipelines increasing output throughput by 45% and reducing inconsistencies by 30%.
- Designed structured data schemas improving dataset quality and cross-team experimentation efficiency by 35%.
- Streamlined data workflows enabling faster iteration and more reliable model outputs.

### Kentush Micro-credit

Mar. 2024 – Jun. 2024

Software Engineer Intern

Kyrgyzstan

- Developed ML credit-risk model preventing 500+ high-risk loans and reducing classification errors by 40%.
- Reduced API latency from 4.5s to 1.8s and improved inference speed by 52% via pipeline optimization.
- Enhanced system performance through efficient data processing and model integration.

### International University of Kyrgyzstan

Jun. 2023 – Mar. 2024

Software Engineer (Part-Time)

Kyrgyzstan

- Built real-time attendance system for 200+ users using FastAPI and event-driven architecture.
- Implemented role-based authentication and optimized SQL queries, reducing admin workload by 30%.
- Automated 95% of attendance tracking using RFID integration and backend optimization.

## Education

### Bellevue College

Jan. 2025 – Jun. 2027

B.S. Computer Science | GPA: 4.0

Bellevue, WA